

Eldridge Surianto

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EDUCATION

Georgia Institute of Technology

Atlanta, Georgia

B.S. in Computer Engineering | Minor in Applications of AI/ML | GPA: 3.95

Expected Graduation: May 2027

- **Concentrations:** Systems and Architecture & Computing Hardware and Emerging Architectures
- **Related Coursework:** Computer Architecture, VLSI & Advanced Digital Design, Semiconductor Physics, Computer Organization, Signal Processing, Circuit Analysis, Digital Design Laboratory, Data Structures & Algorithms, Web Design

SKILLS

Programming: C/C++, Python, Rust, SystemVerilog, Verilog, VHDL, RISC-V, TCL, MIPS, MATLAB, Java, HTML, CSS, Javascript

Technical: FPGAs, HLS, IC Design, UVM, Data Structures & Algorithms, Finite Element Analysis, Web Development, Machine Vision

Tools: Cadence (Virtuoso, Xcelium, SimVision, Verisium), Synopsys (VCS, Verdi), Xilinx (Vitis, Vivado), ModelSim, Quartus Prime, iVerilog, GTKWave, MATLAB, COMSOL, Linux CLI, Git, Oscilloscopes

Professional Organizations: SiliconJackets

Platforms: Linux (Ubuntu, RHEL, WSL), Windows, MacOS

Languages: English, Chinese, German

PROFESSIONAL EXPERIENCE

Software Engineer | LightningX | Sharc Lab | Georgia Tech

August 2025 - Present

- Refactored waveform tracing backend for a **Rust**-based High-Level Synthesis (HLS) simulation framework, reducing execution time by 50% with an optimized waveform output module, enabling efficient reverse engineering of complex FPGA designs.
- Streamlined APATB generation using a **Python** script, leveraging **Clang AST** and **LLVM** bitcode to extract kernel-level code, enabling the creation of automated test benches for validating HLS-generated hardware designs in **Vitis HLS** and **Vivado**.

Software Intern | Automation Technology Center PTE LTD | Singapore

May 2025 - August 2025

- Engineered a high-performance 3D Laser Scanning system in **C++** using **OpenCV** for image processing, **OpenGL** for interactive 3D visualization, and multithreading/**OpenMP** to reduce processing time for 500 images from 60s to <2s.
- Leveraged **Ethernet (SOEM)** and Yaskawa SGD7S servo drive for real-time encoder-triggered camera control via Spinnaker SDK, enabling precise, automated lead frame inspection and integration into semiconductor fabrication machines.

Undergraduate Research Assistant | Plasma & Dielectrics Lab | Georgia Tech

August 2024 - August 2025

- Developed **MATLAB** scripting pipelines to reduce Finite Element Analysis (FEA) setup time in **COMSOL** by over 99%.
- Executed thousands of FEA simulations to analyze impact of material inhomogeneity on current distribution using the Monte Carlo method. Presented poster on research findings and methodology at Lester Eastman Conference 2025.

PROJECTS

RTL Design and Verification of a 64-bit Serial Adder

- Built a 64-bit adder in **SystemVerilog**, iteratively testing core modules like FSM and 32-bit adders using **Xcelium** and **SimVision**.
- Architected an automated, **UVM-style** verification environment in **Synopsys VCS**, leveraging constrained-random stimulus and a self-checking scoreboard with a golden model to achieve over 99% functional coverage in **Synopsys Verdi**.

Pipelined RISC-V Processor

- Implemented a 5-stage pipelined 32-bit RISC-V (RV32I) processor in **SystemVerilog**, implementing hazard detection, stall control, data forwarding, and branch flushing to minimize pipeline bubbles and maximize instruction throughput.
- Debugged and optimized the design with **ModelSim**, **Icarus Verilog**, and **GTKWave**, building custom testbenches to ensure functional correctness, performance efficiency, and compliance with RV32I ISA specifications.

VHDL RTL LED Metronome on DE10-Standard ModelSim-Altera Intel SoC FPGA

- Designed a VHDL-based LED metronome peripheral on a DE10-Standard FPGA using **Quartus Prime** and **ModelSim** timing simulations, incorporating a state machine to control the PWM LED brightness and clock division for adjustable BPM.
- Integrated the peripheral into a custom SCOMP (Simple Computer) architecture, enabling dynamic tempo and LED control via SCOMP assembly instructions directly interfacing with memory-mapped registers.

Wordle Game Embedded Systems Programming on Mbed uLCD kit

- Developed a Wordle-inspired game in C++ for an ARM Cortex-M 32-bit microcontroller, achieving smooth real-time gameplay by optimizing memory usage and code efficiency on a resource-constrained system using GPIO and interrupts.
- Integrated dynamic visuals, audio, and responsive input handling by applying real-time debugging techniques and precise timing control, improving user responsiveness and strengthening hardware-software interaction across all gameplay states.

LEADERSHIP EXPERIENCE

Peer Leader | ECE Discovery Studio | Georgia Tech

August 2025 - Present

- Mentored 15 first-year ECE students in an extended orientation course, providing guidance on academic and social opportunities on campus and organizing activities on professional development like elevator pitches and resume writing.
- Managed course logistics including attendance, discussion, and grading assignments such as resumes and course registration plans; facilitated student project showcase meetings, including a technical article analysis and a personal ECE project.